Meraj Hossain Promit

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Dhaka, Bangladesh

Research Interests

Final-year Robotics & Mechatronics undergraduate working toward **robot learning for locomotion across morphologies**. My current research learns recovery policies for a **wheeled** Mars rover trapped in deformable granular terrain: mobility when contact is uncertain and no hand-designed escape recipe exists. I want to carry that problem from **wheels to quadrupeds to bipeds**, where richer contact and stricter balance make model-based controllers fail more sharply. I am drawn to **reinforcement learning, learning-based control, and Sim2Real**, and to putting simulation-trained policies on hardware. Seeking a **PhD / MS / research position** in robot learning.

Education

University of Dhaka — Dhaka, Bangladesh
B.Sc. in Robotics and Mechatronics Engineering; — August 2022 – October 2026 (Expected)
Fourth Year, Second Semester (8th of 8), Ongoing — CGPA: 3.50 / 4.00

Sylhet MC College — Sylhet, Bangladesh
Higher Secondary Certificate (HSC); — July 2019 – December 2021

Publications

Meraj Hossain Promit, Chandak Chakma, Md. Jubair Ahmed Sourov, and Sejuti Rahman. "Escape Without a Recipe: Maneuver-Agnostic Mars Rover Recovery from Granular Entrapment." *Space Robotics Workshop* at IEEE/RSJ IROS 2026 (Pittsburgh, PA). **Accepted**, poster. [Workshop]

Motivation: Wheel entrapment in loose regolith has crippled real planetary missions (e.g., NASA's Spirit rover). We learn a maneuver-agnostic recovery policy in deformable granular dynamics, restoring mobility without a prescribed escape recipe.

Research Experience

MARVEL Lab, University of Dhaka — Dhaka, Bangladesh
Undergraduate Researcher, Lab Page — April 2025 – Present

Supervisor: Dr. Sejuti Rahman, Associate Professor (on leave), University of Dhaka; Professor of CS, New Uzbekistan University.

- **Lead author** on DRL controllers that recover rover mobility in deformable Mars regolith, learning locomotion policies where classical control fails under granular slip.
- Designed **yaw-projected reward shaping with entrapment-gated slip penalties** (a safety-style constraint) and built large-scale, multi-GPU training and evaluation pipelines.

University of Texas Rio Grande Valley — Rio Grande Valley, TX, USA (Remote)
Research Assistant — August 2025 – Present

- Building a digital-twin warehouse benchmark to evaluate Deep RL navigation policies (PPO, SAC, TD3) under realistic perception and localization noise. [GitHub]

North South University — Dhaka, Bangladesh
Undergraduate Research Student — January 2024 – December 2024

- Studied melanoma segmentation from pseudo-annotations under limited labels (weakly-supervised computer vision). **Supervised by** Dr. Mahdy Rahman Chowdhury, Associate Professor, ECE, NSU.

Projects

ALBATROSS: Autonomous Maritime Patrol UAV | PX4, Gazebo, ROS2, MAVLink — [GitHub]

- Built a software-in-the-loop maritime-patrol drone over real Chattogram port terrain: PX4 SITL and Gazebo for flight, with a ROS 2 A* planner that routes exclusion-aware waypoints and never enters a no-fly zone.
- Shipped a browser ground station with live 2D/3D maps, a multi-waypoint mission editor (per-waypoint altitude), MAVLink control, and a Gazebo chase-camera feed.

Manipulator Kinematics & Dynamics from Scratch | NumPy, ROS2, PyBullet [GitHub]

- Derived a full kinematics/dynamics stack for a 6-DOF UR5e and 7-DOF KUKA iiwa with no MoveIt, KDL, or Pinocchio; validated against official URDFs to 1.5×10^{-7} and caught a DH sign error that left the UR5e model 1.998 m off.
- Implemented damped-least-squares IK, quintic Cartesian blending with quaternion slerp, and Newton–Euler vs. Lagrangian dynamics (agree to 1.6×10^{-11} Nm); 183 tests.

RRT* Motion Planning in 7-DOF Configuration Space | NumPy, PyBullet [GitHub]

- Wrote asymptotically optimal RRT* from first principles for a redundant iiwa carrying a mug past a non-convex mesh obstacle, checking the payload as moving geometry; converges to within 0.16% of the analytic optimum on a 2-D validation case.

STM32 Quadcopter Flight Controller | STM32, Embedded C, Sensor Fusion [GitHub]

- Bare-metal STM32F103 controller at 250 Hz with no RTOS: complementary-filter fusion of IMU, magnetometer, barometer, and GPS, closing six loops (rate, attitude, altitude, GPS hold). Built in 2023 and flown, re-tuned, and upgraded since.

WarehouseBenchmark | Deep RL, NVIDIA Isaac Lab, ROS2 [GitHub]

- Engineered a 31.8×54 m digital-twin warehouse benchmarking PPO, SAC, and TD3 on a Clearpath Jackal with realistic LiDAR and localization noise; curriculum learning and domain randomization for sim-to-real robustness.

Technical Skills

Reinforcement & Robot Learning: PyTorch, NVIDIA Isaac Lab, PPO / SAC / TD3, reward shaping, domain randomization, curriculum learning, Sim2Real

Robotics & Simulation: ROS2, Gazebo, PX4, MAVLink, Nav2, MoveIt, SLAM, numerical IK, RRT*, OpenCV

Programming & Systems: Python, C / C++ (learning CUDA), Bash; Linux (Ubuntu, Arch, Kali); Git

Hardware & CAD: STM32, Raspberry Pi, Fusion 360, ANSYS, PCB design

Leadership & Engineering Experience

Intelis Solution Ltd. Dhaka, Bangladesh
Embedded Engineer (Part-time) July 2026 – Present

- Building STM32-based autonomous drone platforms for maritime monitoring, with robust firmware, hardened telemetry links, and weather-resistant sensing for harsh sea conditions.

Karthikesh Robotics Pvt. Ltd. Chennai, India (Remote)
ROS Development Intern June 2025 – Present

- Developing autonomous robots for industrial and educational use with ROS2 navigation and perception stacks.

Mohakashchari, Dhaka University Rover Team Dhaka, Bangladesh
Project Lead May 2025 – Present

- Proposed the concept at IEEE SB DU; now leading design, frame integration, and fundraising for a competition-ready rover platform.

IEEE Robotics & Automation Society, University of Dhaka Dhaka, Bangladesh
Chair (2025–Present); previously Student Activity Secretary (2023–2024)

- Lead chapter activities, events, and research collaborations promoting robotics and automation among students.

Awards & Achievements

8th Place, DU AI Challenge (2025): computer-vision competition among university teams.

4th Place, Bangladesh Olympiad on Astronomy & Astrophysics (2021): national winner and training-camp participant.

Rank 62, HSC 2021 (Sylhet Board): top 100 across all examinees in the board.

Relevant Coursework

Robotics & AI: Artificial Intelligence, Intelligent Systems & Robotics, Advanced Robotics, Control Systems Design, Digital Image Processing & Robot Vision, Digital Signal Processing, Mechatronics (Intro & Advanced)

Math & CS: Linear Algebra, Multivariable & Vector Calculus, Differential Equations, Numerical Analysis, Probability & Statistics, Data Structures & Algorithms, Object-Oriented Programming